

Keep the Angle

A Geometry-Preserving Basis in Spectral Embeddings

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Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances are known to *degenerate* on large graphs [14]. We show the geometry does not vanish—it migrates entirely into the *angular* coordinate. Writing each node’s low-mode embedding in polar form (magnitude $\times$ direction), the direction preserves graph geodesics at Spearman $\rho \approx 0.93$ on a reference manifold, while the magnitude preserves $\rho \approx 0.03$ and a random-mode basis of equal size preserves $\rho \approx 0$. Keeping only the angle is exactly the row-normalization of Ng–Jordan–Weiss spectral clustering [8], whose existing theory concerns cluster separation [12]; we give it a geodesic justification. We show, as a corollary of von Luxburg–Radl–Hein, that the radial coordinate of the full embedding degenerates to a local-degree quantity for intrinsic dimension $d \geq 2$, and **conjecture**, with strong empirical support, that the angular coordinate **remains rank-faithful to geodesic distance**—flat in both mode-count and graph size where the raw embedding decays. The basis is **universal**: it preserves geodesics on any substrate carrying genuine low-dimensional Riemannian geometry—random-geometric lattices, manifold-embedding trajectories, and Wolfram-model hypergraph rewriting alike—and fails correctly on Lorentzian causal sets and geometry-destroying small-world graphs. Across 20 independent emergent manifolds ($d \approx 1\text{--}3.6$), fidelity falls approximately linearly with intrinsic dimension. Refined toward the continuum (N up to 5×10^4), the basis *sharpens* on sampled Riemannian substrates (angular-fidelity Spearman $+1.00$ with $\log N$ at $d=3$) yet *decays* below a pre-registered floor on a coordinate-free Wolfram-model manifold—a dissociation that recasts the angular coordinate as a *diagnostic* for whether a substrate’s fine structure stays manifold-like under refinement.

1 Introduction

Spectral embeddings—Laplacian eigenmaps [1], diffusion and commute-time maps [5]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [14]. Yet the most successful spectral-clustering algorithm [8] *row-normalizes* the embedding—projects each node to the unit sphere—and works well. The existing theory of that normalization concerns cluster geometry: it explains why row-normalized embeddings of well-separated clusters concentrate in nearly orthogonal directions on the sphere [12, 13]. It does not address metric geometry. **This paper asks what row-normalization recovers of the graph’s geodesic structure, and why.**

Our answer is a clean decomposition. Write each node’s low-mode embedding in polar form, magnitude $\times$ direction. The *magnitude* is exactly the degenerate radial coordinate that von Luxburg’s

theorem kills; the *direction* carries the graph’s geodesic geometry. Row-normalization keeps the direction and discards the magnitude—so beyond its cluster-separation role it is also the geodesically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

This “keep the scale-invariant direction” operation recurs beyond clustering—for example in polar-coordinate KV-cache quantization, which transfers the scale-invariant direction and quantizes the radius separately [10]. It also admits a physical reading: in observer theory a bounded observer perceives geometry only after coarse-graining a discrete substrate [17], and the angular projection is a candidate for that coarse-graining. We keep that interpretation to the discussion (§7) and let the spectral result stand on its own; the empirical core is that the basis is *universal*, preserving geodesics on any graph carrying genuine low-dimensional Riemannian geometry, independent of how it was generated.

Contributions: (1) the **Keep-the-Angle Principle** (§3)—a Radial-Degeneracy *theorem* (for the full embedding, as a corollary of [14]) and an empirically-supported Angular-Preservation *conjecture*; (2) a measurement with controls (§5); (3) universality across five substrate types (§5.3); (4) a fidelity-vs-dimension scaling law (§5.4); and (5) two honest negatives (§6). We do *not* claim to derive physics; the core result is a basis-level fact about spectral embeddings.

2 Background

Spectral geometry. Laplacian eigenmaps [1] and diffusion maps [5] embed a graph via low eigenvectors; the commute-time embedding scales mode k by $1/\sqrt{\lambda_k}$. **Commute-time degeneracy.** von Luxburg et al. [14] proved the commute distance of a large geometric graph degenerates, $R(i, j) \rightarrow 1/k_i + 1/k_j$ (k_i the degree of node i); we read this backwards as the collapse of the *radial* coordinate to local degree. **Row-normalization.** Ng–Jordan–Weiss [8] row-normalize the eigen-embedding—the angular projection we study. Schiebinger, Wainwright and Yu [12] give the sharpest existing account: under a mixture model, row-normalized embeddings concentrate clusters in orthogonal directions, explaining k -means’ post-normalization success; consistency of spectral clustering is treated in [13]. Both analyses are about cluster separation on the sphere. Our question—whether the normalized embedding preserves *geodesic distances* on a manifold-like graph—is complementary, and to our knowledge open. **Emergent geometry and observer theory.** In the Wolfram model [15, 16, 6] space is a hypergraph and law is what a bounded observer perceives after coarse-graining the ruliad [17]; that coarse-graining is left qualitative, and our angular projection is a candidate for it (§7). **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [7, 11, 9]; we test and reject a hyperbolic reframe in §6.

3 The Keep-the-Angle Principle

Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k) , $0 = \lambda_0 < \lambda_1 \leq \dots$. The **low-mode spectral** embedding using the lowest m nontrivial modes places node i at $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$. Write $\Psi_i = r_i \hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$.¹

¹We call Ψ “commute-style” rather than the commute-time embedding proper: the classical commute embedding uses the pseudoinverse of the *unnormalized* Laplacian and all $n - 1$ modes, so that $\|\Psi_i - \Psi_j\|^2$ equals the resistance $R(i, j)$ exactly. Our Ψ is the truncated, normalized analogue standard in spectral clustering; the distinction matters for Corollary 1 and is handled there.

Intrinsic dimension and degree notation. Throughout, d is the intrinsic dimension of the substrate, and k_i denotes the degree of node i (we reserve d for dimension to avoid the collision d vs. d_i): the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices, sphere), and for emergent graphs the value at which the three independent estimators of §4 (ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. The regime $d \geq 2$ refers to this measured intrinsic dimension; §5.4 confirms the $d = 1$ boundary behaves as predicted.

Corollary 1 (Radial Degeneracy). *Let G_n be random geometric graphs of intrinsic dimension $d \geq 2$ satisfying the conditions of [14], and let Φ_i be the full commute-time embedding built from the unnormalized Laplacian, so that $\|\Phi_i\|^2 = L_{ii}^+$. As $n \rightarrow \infty$ the resistance distance degenerates, $R(i, j) \rightarrow 1/k_i + 1/k_j$; since the only per-node content of this additively separable form is the self-term L_{ii}^+ , the radius satisfies $\|\Phi_i\| \rightarrow 1/\sqrt{k_i}$. The radius of the full commute embedding is therefore asymptotically geometry-free—a local-density coordinate carrying no geodesic information.*

Remark (Truncated, normalized radius). Our experiments use the truncated radius $r_i = \|\Psi_i\|$ (m lowest modes of L_{sym}), not $\|\Phi_i\|$. Transferring Corollary 1 to this setting requires (i) the normalized analogue of the degeneracy in [14] and (ii) control of the truncation error, and we do not prove either here; we state the truncated statement as an empirical observation. Numerically it holds tightly: $\rho(r_i, 1/\sqrt{k_i}) = 0.92\text{--}0.99$ and radius-only geodesic preservation $\rho \leq 0.08$ (§5.5).

Conjecture 1 (Angular Preservation). *Row-normalization $\hat{u}_i = \Psi_i/\|\Psi_i\|$ deletes the degenerate density factor of Corollary 1, and the pairwise distances among the resulting directions remain rank-faithful to geodesic distance—Spearman correlation bounded away from zero (empirically ≈ 0.9)—uniformly in the mode-count m and size n . A stronger, metric form of the conjecture—that the angular map is bi-Lipschitz to geodesic distance with constants uniform in m and n —is consistent with our data but untested; our evidence is rank-based and cannot by itself bound metric distortion.*

Status. Empirically strong (§5) but **not proven**. It rests on eigenmap-embedding results plus a heuristic; the honest gap is that Corollary 1 explains why the radius *dies* for $d \geq 2$, not by itself why the angle *lives*—indeed the angular fidelity persists ($\rho \approx 0.88\text{--}0.90$) even on quasi-1D graphs where Corollary 1 does not apply, so the angular half is carried by the eigenmap-embedding mechanism, not degeneracy alone. That mechanism is analyzable rather than mysterious: if the low L_{sym} eigenvectors converge to Laplace–Beltrami eigenfunctions [2, 3] and eigenfunction coordinates furnish locally bi-Lipschitz charts of the underlying manifold [4], the angular map’s geodesic fidelity should follow from the chart geometry—the natural route to a proof of Conjecture 1, and this paper’s central open problem. Together these are the **Keep-the-Angle Principle**: keep $\hat{u}_i$, discard r_i .

Dimension gating. Corollary 1 is a $d \geq 2$ phenomenon. In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Corollary 1 correctly does not apply—a prediction, not a failure, confirmed in §5.5.

4 Methods

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread $\Rightarrow$ clean manifold, and the agreed value is our measured intrinsic dimension d . **Angular metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (unit-normalized low-mode embedding), *random* (m random modes; control), *magnitude* (radius

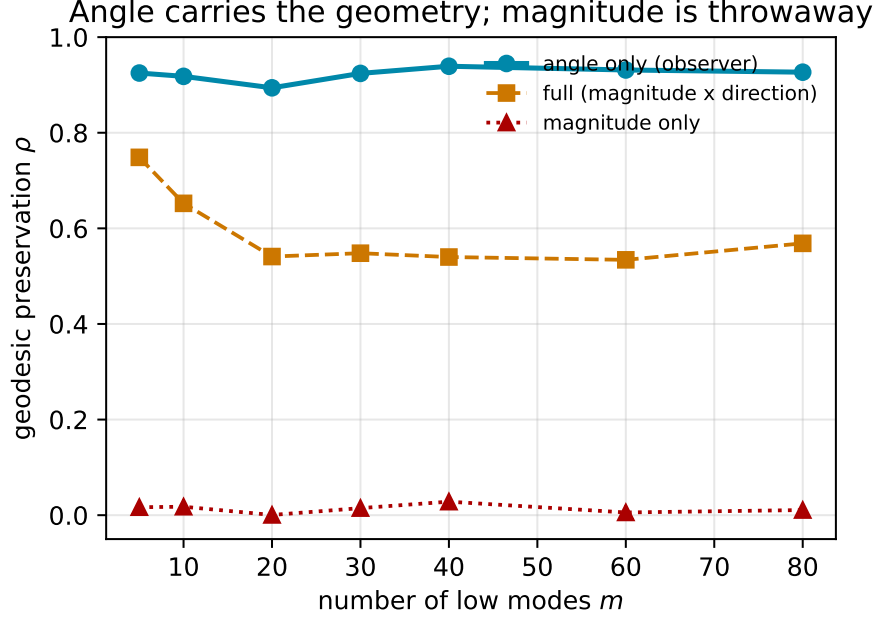


Figure 1: Spearman correlation with graph geodesics on a 2-torus ($n = [\text{TODO}]$) as a function of mode-count m . Angle carries the geometry; magnitude is throwaway; the full commute embedding decays with m .

only), *euclid* (classical 2D MDS). **Uncertainty.** [TODO: report, for every ρ in the paper, the number of sampled node pairs, and bootstrap 95% confidence intervals over pair resampling; for headline numbers, also variation over independent graph draws.] **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding trajectories. [TODO: specify what the swiss-roll “trajectory” data actually is—source, length, and how the graph is built from the sequence—since §5.3 and §7 lean on it as the sequential-data case.]

5 Results

5.1 The core result

Estimators recover ground-truth dimension within ≈ 0.3 . On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

m	full (mag. $\times$ dir.)	angle only	magnitude only
5	0.761	0.926	0.025
10	0.650	0.913	0.035
20	0.550	0.891	0.028
40	0.517	0.928	0.034

The angle carries the geometry (≈ 0.92 , flat in m); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 2).

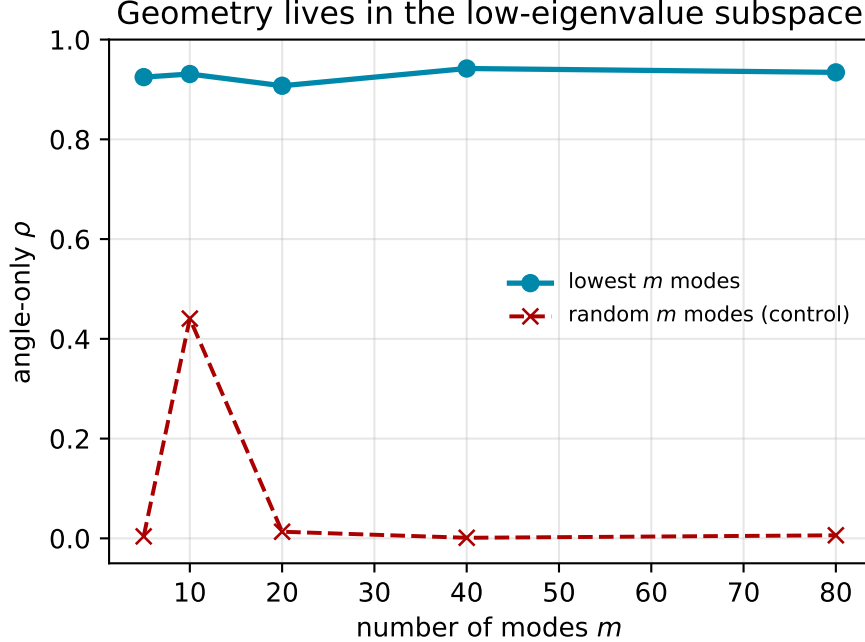


Figure 2: Geometry lives in the low-eigenvalue subspace: lowest- m modes preserve geodesics; a random-mode basis of equal size does not.

5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only $\approx 1.5D$ manifolds (82 manifold-grade rules in a 20k sweep; none at $d \geq 2.5$). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020 rule ($d = 2.07 \pm 0.01$, spread 0.05, angle- $\rho = 0.82$) and by from-scratch evolutionary search (≈ 3 generations). The angular basis holds on genuine 2D emergent space.

5.3 Universality across substrates

substrate	meas. d	angle- ρ	random	euclid
2-torus (reference)	2	0.89	0.01	0.67
king-lattice	2	0.82	0.01	0.95
swiss-roll trajectory	2	0.93	0.01	0.99
emergent R_{3D} (Wolfram)	2.07	0.82	0.01	0.81
causal set (Lorentzian)	3.9	0.57	0.01	0.55
small-world (destroyed)	3.0*	0.45	0.00	—

Substrates with genuine low- d Riemannian geometry all show the effect, independent of graph origin; the metric fails correctly where geometry is Lorentzian or destroyed. *For the small-world graph the three dimension estimators disagree [TODO: report the inter-estimator spread]; by our own manifold-grade criterion this graph is *not* a manifold, and the low angle- ρ is the predicted failure, not an exception.

Two caveats keep the universality claim honest. First, on the two flat substrates (king-lattice, swiss-roll) classical MDS *beats* the angular basis (0.95 and 0.99 vs. 0.82 and 0.93): MDS wins whenever the manifold embeds near-isometrically in low-dimensional Euclidean space, while the

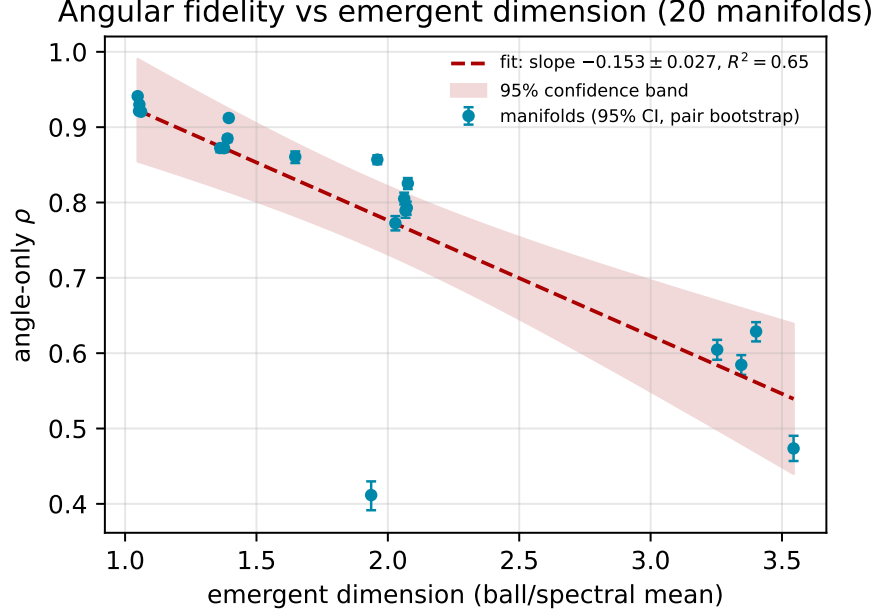


Figure 3: Angular fidelity falls linearly with emergent dimension across ring, tori, and Wolfram-model rules. [TODO: add a confidence band on the fit and per-point error bars from pair-resampling.]

angle wins where topology obstructs such an embedding (the torus: 0.89 vs. 0.67) and, unlike MDS, is uniform in m and n (§5.5). Universality here means the angular basis never *loses* the geometry on a genuine low- d Riemannian substrate—not that it dominates every baseline on every substrate. Second, the swiss-roll trajectory is the sequential-embedding case: the angular basis is the same whether the manifold comes from hypergraph rewriting or sequential trajectory data.

5.4 The scaling law

Across 20 independent emergent manifolds ($d \approx 1\text{--}3.6$),

$$\text{angle-}\rho \approx 1.09 - 0.157d, \quad \text{slope } -0.157 \pm [\text{TODO: SE}], \quad R^2 = [\text{TODO}], \quad d \in [1.0, 3.6].$$

(Spearman -0.881 between ρ and d serves as a distribution-free monotonicity check; the linear fit itself should be judged by its slope uncertainty and R^2 .) The fit is *empirical and local*: ρ is a rank correlation bounded by 1, so the linear form holds only within the measured range—the intercept 1.09 is an extrapolation, not a literal $d \rightarrow 0$ fidelity, and the slope may vary with substrate roughness. Within range, fidelity is highest at low dimension and softens toward ≈ 0.56 at $d \approx 3.4$ (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism.

5.5 Verifying Corollary 1 and Conjecture 1

angle- ρ is flat in graph size while the full commute distance decays, and the radius tracks degree exactly ($\rho(\|\Psi_i\|, 1/\sqrt{k_i}) = 0.92\text{--}0.99$; radius-only $\rho \leq 0.08$). Corollary 1 holds cleanly at $d = 3$ ($\rho(R, 1/k_i + 1/k_j) \rightarrow 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 1: in an m -sweep at $n = 4000$, angle- ρ is flat ≈ 0.93 across $m = 5\text{--}80$ while the full embedding decays $0.773 \rightarrow 0.437$.

5.6 Stability across substrate evolution

Universality across substrate *types* (§5) is a statement about fixed graphs. The angular basis also persists as a substrate *evolves*. For each of 151 qualifying Wolfram-model rules (95 arity-2, 56 arity-3) we embed successive rewrite snapshots on a fixed anchor set and compare, between consecutive snapshots, the Spearman persistence of the *angular* pairwise structure (**consec**) against the persistence of the underlying *geodesic* structure itself (**churn**)—the substrate’s own rate of change is the baseline the basis must keep pace with. On all 143 rules resolvable at protocol scale (8 never yield ≥ 3 distinct snapshots and are excluded from the denominator but reported), the angular structure tracks the evolving geometry at least as stably as the graph’s own geodesics, $\overline{\text{consec}} \geq 0.9 \overline{\text{churn}}$: a 143/143 = 100% pass against a pre-registered 80% bar, with zero substrate-unstable rules. Over the longest time-lever (first vs. last snapshot) angular persistence declines only gently with emergent dimension,

$$\text{slope} = -0.029/\text{dim}, \quad \text{Spearman} = -0.48, \quad d \in [0.98, 2.23],$$

and this replicates independently in both arities (arity-2 $-0.028/\text{dim}$, $\rho = -0.45$, $n = 95$; arity-3 $-0.029/\text{dim}$, $\rho = -0.54$, $n = 48$)—the same mechanism, unchanged by moving from binary to ternary rewriting.²

5.7 Continuum survival, and a dissociation

Every result so far is at finite N ($\approx 2\text{--}4\text{k}$ nodes, capped by dense **eigh**); the “uniform in N ” clause of Corollary 1 and Conjecture 1 is untested at scale (§5.5). We test it directly by refining each substrate toward the continuum with a sparse eigensolver, $N \in \{2, 5, 10, 20, 50\} \times 10^3$, four seeds, $m = 10$ modes, $n = 300$ anchors—well past the $\sim 5\text{k}$ dense ceiling. The two substrate families dissociate. On genuine Riemannian substrates the basis *sharpens*: on random geometric graphs the angular fidelity is flat-to-rising in N , most cleanly in the higher-dimensional case,

$$\begin{aligned} \text{RGG } d=3 : \quad \text{angle-}\rho : 0.856 \rightarrow 0.912 \quad (N : 2\text{k} \rightarrow 50\text{k}), \\ \text{Spearman}(\text{angle-}\rho, \log N) = +1.00, \end{aligned}$$

with $d=2$ likewise monotone ($+0.90, 0.916 \rightarrow 0.929$): refining the mesh makes the low modes resolve the geometry *better*, not worse, and both controls stay pinned (higher-mode $\rho \leq 0.05$, magnitude $\rho \leq 0.03$, eigensolver residual $\sim 10^{-14}$) at every N . The finite- N evidence of §§5–5.5 was not a small-graph artifact.

The dissociation is with *emergent* substrates. The $\text{R}_{3\text{D}}$ Wolfram-model manifold—geometry with no latent coordinates—*decays* monotonically with refinement, $\text{angle-}\rho : 0.749 \rightarrow 0.654$ (Spearman = -1.00), falling below a pre-registered floor of 0.75; and at the largest N its magnitude control begins to break the clean angle/radius separation that holds exactly on the RGGs (one 50k seed crosses $\rho_{\text{mag}} = 0.33$ and is gated out). A sampled manifold refines to a clean continuum; an emergent hypergraph, under this basis, does not—its fine structure drifts away from a manifold as it grows. The coordinate-free substrate—the one the observer-theory reading most needs—is exactly the one where the basis fails in the limit. We read this not as a limitation but as a result: the angular basis is a *diagnostic* for whether a substrate’s fine structure stays manifold-like under refinement, and it returns different verdicts for sampled (2/3 pass) and emergent (1/1 fail) geometry.³

²Pre-registered as Amendment 2: an earlier capture hook could append the same state once per crossed edge-count threshold, inflating consecutive stability toward a spurious 1.0. The corrected rule takes at most one snapshot per generation with dynamic $2\times$ spacing; the numbers here are from that harness.

³Pre-registered (PREREG_RUNG4) before the run; survival bar $\text{angle-}\rho(N_{\text{max}}) \geq \max(0.75, 0.9 \text{angle-}\rho(N_{\text{min}}))$

6 Honest Negatives

Not secretly hyperbolic (rejected). With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature is not dynamical (null).** Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw -0.98 against update density that deflates to a tautology (update-count equals degree, $\rho = +1.00$; hubs are negatively curved); controlling for degree, the independent signal is partial- $\rho = -0.14$. This *bounds the observer-theory interpretation*—no emergent Einstein tensor—and leaves the spectral result untouched.

7 Discussion and Conclusion

The primary result is basis-level and substrate-agnostic. The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is von-Luxburg-degenerate noise. This gives Ng–Jordan–Weiss row-normalization a *geodesic justification complementary to the cluster-separation theory of [12]*, unifies it with the scale-invariant direction of KV-cache compression [10], and holds *universally* across substrates with genuine low- d Riemannian geometry, independent of graph origin. That the same basis works identically on hypergraph-rewriting and *sequential-trajectory* substrates is *evidence that scale-invariant angular coarse-graining is a general mechanism by which* relational data carries geometric structure.

One interpretation is observer-theoretic. In Wolfram’s observer-relative framework, the coarse-graining a bounded observer performs to perceive geometry is exactly this angular projection—making “the observer’s basis” a concrete candidate for the coarse-graining the program leaves unspecified. We offer this as motivation and bound it honestly: its strongest claim—dynamical, energy-responsive curvature—is a null (§6). The core contribution stands independent of the physics reading. Open problems: proving Conjecture 1 (in its rank form, and ideally the bi-Lipschitz strengthening); extending Corollary 1 to the truncated, normalized radius actually used in practice; synthesizing clean manifolds above $d \approx 2$; and a dimension-adaptive angular basis that tunes mode-count and diffusion time to counter the fidelity decay. Keep the angle, drop the magnitude: the basis-level fact is the contribution; the observer interpretation is where we leave the door open.

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on $\geq 2/3$ substrates. Amendment 1, registered before any run: the whole-spectrum random-mode control needs a dense decomposition unavailable at $N \geq 50k$, so it is the top m of the lowest-64 sparse-computed modes—a faithful, sparse-affordable stand-in, same ≤ 0.15 bar.

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